Veer Chheda

Mumbai, India

+91 8451820598 | veerchheda@gmail.com | linkedin.com/in/veerchheda | github.com/veer-chheda

Education

Dwarkadas J. Sanghvi College of Engineering — **Nov 2022 – Jun 2026**

B.Tech in Computer Science and Engineering (IoT and Cybersecurity with Blockchain Technology) — *CGPA 9.12/10 - Top 10*

Relevant Coursework (Language of Instruction) - English

Data Structures & Algorithms, Operating Systems, Machine Learning, Artificial Intelligence, Deep Learning

Experience

Indian Institute of Technology (IIT) Patna — **Aug 2024 – Nov 2025**

Research Intern — *Remote*

- Conducting research under **Dr. Rajiv Misra** and**Dr. Sourav Dandapat**, focusing on NLP and computer vision.
- Developed mechanisms to identify and mitigate **hallucinations** in large language models for **MedQA** datasets using **ground-truth similarity** and **knowledge-base injection** for mitigation.
- Trained and optimized custom **NanoDet** and **YOLO** models for robust **small object detection** under varied conditions.
- Fine-tuned **LlaVA-next-video, Qwen 2.5 VL, Phi-4 mini** on multimodal summarization of car crash videos using both textual and visual data.

Aura IT (AURA-IT Tech Solutions Pvt. Ltd.) — **Jun 2024 – Oct 2024**

Data Science Intern — *Remote*

- Curated footfall analytics solutions, including **heatmaps** and **re-identification** using YOLO, achieving 86% accuracy in identifying customer retention patterns and hotspots, boosting brand visibility by 25%.
- Integrated **OCR** models using **Amazon Textract** into commercial pipelines for **automated data entry**, reducing manual processing time by 47%.
- Collaborated with a **cross-functional team** of 5+ members to design and deploy APIs using **FastAPI** and secure authentication using **JWT**.

Research and Publications

Extract-Explain-Abstract: A Rhetorical Role-Driven Domain-Specific Summarisation Framework for Indian Legal Documents — *First Author* | Paper

- Published at the **7th Natural Legal Language Processing Workshop**, **EMNLP 2025**.
- Developed the **Extract-Explain-Abstract** (EEA) framework for summarizing Indian legal documents via rhetorical role segmentation.
- Modeled logical dependencies between roles using chain-of-thought, cutting hallucinations by **17%**.
- Evaluated on **ILC** and **IN-Abs** datasets with ROUGE, BERTScore, InLegalBERTScore, Flesch Reading Ease and SummaC.
- Human and expert evaluations showed **33%** preference and **40%** gold-standard replacement endorsement over baselines.

Iterative Critique-Driven Text Simplification: Targeted Enhancement of Complex Definitions with Small Language Models — *First Author* | Accepted

- To be published in **Findings of IJCNLP-AACL 2025**.
- Proposed an iterative zero-shot critique-driven framework for simplifying domain-specific definitions using small language models, driven by cognitive accessibility.
- Used LLM critics to iteratively refine definitions for lay audiences, optimizing clarity, precision, conciseness and relevance.
- Benchmarked on **WIKIDOMAINS** dataset with 22k+ definitions, showing gains in BLEU-4, BERTScore, Age-of-Acquisition and Flesch Reading Ease.
- Human evaluation (**18 annotators**) and **LLM-as-a-judge** confirmed significant comprehension gains, especially for non-experts, via personalized refinement.

Could you BE more sarcastic? A Cognitive Approach to Bidirectional Sarcasm Understanding in Language Models — *First Author* | Accepted

- To be published at the **Student Research Workshop**, **IJCNLP-AACL 2025**.

- Introduced a multi-hop chain-of-thought framework for bidirectional sarcasm style transfer, leveraging Theory of Mind.
- Benchmarked 6 SLMs on sarcasm generation and removal using the **MUStARD** dataset (690 dialogues).
- Evaluated with automatic (BLEU, DialogRPT, SBERT, BERTScore), LLM-as-a-judge and human judgements (5 annotators, 60 cases).
- Improved metrics by up to 12% over few-shot baselines and won top human rankings across all four style directions.
- Ranked best in all 4 tasks, with Krippendorff's alpha 0.45, excelling in context, creativity, meaning preservation and sarcasticness.

Color Space Fusion for Semantic Segmentation in Indian Adverse Weather Driving Scenes *First Author* | Accepted

- Accepted at the **6th Biennial International Conference on Nascent Technologies in Engineering** to be published in **IEEE Xplore**.
- Proposed and benchmarked 4 color space fusion methods (**YUV, LAB, HSV, direct RGB baseline**) for integrating RGB-NIR images on the **IDD-AW** dataset with **5,000 RGB-NIR image pairs** covering four weather conditions (**rain, fog, low-light, snow**).
- LAB and YUV fusion yielded the greatest gains in fog and low-light (**mIoU up to 30.95**, **SmIoU up to 22.28**) with all fusion approaches preserving real-time inference on edge hardware.

Evaluating Linguistic Proximity of Marathi as a Bridge Language for Konkani Translation *First Author* | Accepted

- Accepted at the **6th Biennial International Conference on Nascent Technologies in Engineering** to be published in **IEEE Xplore**.
- Evaluated zero-shot and fine-tuned English-to-Konkani translation using **IndicTrans2** on parallel translation dataset and Swadesh Lists.
- Achieved highest performance among Devanagari-script Indo-Aryan languages with fine-tuned Marathi translation scores: BLEU-4 (**32.95**), TER (**43.21**), chrF3 (**72.02**), BERTScore (**0.888**) and semantic similarity (**0.868**).

Predicting Personality from Text: A Dual-Task Psycholinguistic Analysis of Machine Learning and Transformer Models *Second Author* | Under review

- Under review at the **8th IEEE International Conference on Emerging Smart Computing and Informatics**.
- Performed dual-task evaluation of MBTI personality prediction on the MBTIBench dataset with **286 text samples** using multi-label classification and multi-output regression with traditional ensemble ML models (**XGBoost, Random Forest, SVM**) and transformer models (**BERT, RoBERTa, DeBERTa**).
- Conducted interpretable psycholinguistic analysis with **SHAP** and **Empath** lexicon, identifying lexical cues aligned with MBTI dimensions and highlighting distinct linguistic patterns for each personality trait.

Projects

Deep Face Detection | *OpenCV, Tensorflow, Keras* **Github**

- Formulated a custom CNN architecture using transfer learning and weights from VGG-16 with 2 chain layers–one for classification and one for bounding box extraction, achieved 96% accuracy.
- Assembled Labelme for data annotation and albumentations for image augmentation pipelines for increased efficiency.
- Deployed the model using Streamlit using WebRTC.

JainGPT - AI chatbot for Jainism | *Google Cloud, LangChain, ChromaDB* **Github**

- Designed a RAG application for custom knowledge chatbot using over 20 scriptures and books on Jainism.
- Combined ChromaDB and VertexAI to generate and store vector embeddings.
- Programmed prompt engineering for text retrieval and text translation into Gujarati, text-to-speech using Google Cloud APIs.

Technical Skills

Languages: Python, C++, C, HTML/CSS, JavaScript, SQL
Frameworks/Libraries: React.js, Next.js, NumPy, Pandas, PyTorch, FastAPI, OpenCV, Unsloth
Tools: Git, GitHub, Firebase, MySQL, LabelStudio

Leadership / Extracurricular

DJS CodeStars **Jul 2023 – Current**
Vice Chairperson (Technical) *Dwarkadas J. Sanghvi College of Engineering*

- Administered one of the most regionally prominent intercollegiate competitive programming event in India - Code Uncode with over 2000 registrations.
- Managed the competitive programming community in the university with timely contests, lectures and sessions by alumni.